# F39 — CKM color$^2$ Direction A is disfavored, Direction B favored (color-counting); corrects F36 lean

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## F39 CKM color$^2$ — Direction A disfavored, B favored

### 0. Goal

F36 named two directions for the substrate-color$^2$ source in $\sin\theta$_C = N_c$^2$/(2^{N_c}·n_C) = 9/40 = $\lambda$, and tentatively leaned toward Direction A (the forced primitive is the rate $\sin^2\theta$_C, where N_c$^2$ is "natural as a two-trace"). Pulling Direction A at depth shows that lean was wrong. This corrects it.

### 1. Direction A does not resolve color$^2$ — it relocates it

The claim was: if the substrate forces the *rate* $\sin^2\theta$_C rather than the amplitude, then N_c$^2$ is the natural two-color-trace factor of |amplitude|$^2$. Check the color powers:

- amplitude $\sin\theta$_C = N_c$^2$/(2^{N_c}·n_C) = 9/40 — carries N_c$^2$ (two color powers).
- rate $\sin^2\theta$_C = N_c$^4$/(2^{N_c}·n_C)$^2$ = 81/1600 = $\lambda^2$ — carries N_c$^4$ (four color powers).

A rate carries two color traces (|amp|$^2$). So a rate with N_c$^4$ requires the *amplitude* to carry N_c$^2$ — which is exactly the thing we were trying to explain. Direction A therefore does not make N_c$^2$ natural; it pushes the question from "why N_c$^2$ in the amplitude" to "why N_c$^4$ in the rate = (N_c$^2$ amplitude)$^2$." Same puzzle, relocated. Direction A is disfavored.

### 2. Direction B is the candidate, and F24's "3" doesn't refute it

Direction B: the amplitude carries N_c$^2$ because the substrate mixing matrix element ⟨V_{gen-1}⊗color | T_W | V_{gen-2}⊗color⟩ has two independent color sums — the gen-1 (down) quark's color triplet and the gen-2 (strange) quark's color triplet

— summed independently in the Bergman inner product, rather than tied by a single $\delta$ as in the SM color-diagonal single-trace picture.

F24 v0.1 reported a color factor of 3 (not 9) and seemed to block Direction B. But F24 computed the Sym^k symmetric-power index combinatorics (1×3 across the gen-1→gen-2 K-types), which is the *representation degree* structure — not the color-triplet index sums. Those are different objects: the Sym^k degree is the Sp(2) symmetric-power count (Ch 7); the color sum is over the SU(N_c=3) triplet attached to each quark. F24's "3" is a category mismatch with the color$^2$ question and does not refute Direction B.

## 3. The sharpened test

The question is now single and concrete: in the substrate Bergman matrix element for d→s mixing, are the initial-quark and final-quark color sums independent (⍰ N_c × N_c = N_c$^2$ in the amplitude, Direction B confirmed) or tied by color conservation to a single trace (⍰ N_c, and 9/40 is not color$^2$)?

- If independent: N_c$^2$ in the amplitude is forced, sin $\theta$_C = N_c$^2$/(2^{N_c}·n_C) = $\lambda$ is a genuine substrate-mechanism form, and the color-diagonal-single-trace intuition simply fails because the two quarks live in *different* generation K-types whose color indices are not identified.
- If tied: 9/40 = $\lambda$ remains an excellent identification (0.000%) but the "9" is not color$^2$ — it is one of the Casey #5 composites (e.g. 2^{N_c}+1, or C_2+N_c), and the color reading is retired.

This is a clean falsifiable fork, resolvable by the explicit FK Bergman color-tensored matrix element (the same Ch 7 machinery F38 needs for $\varepsilon$). One computation decides it.

## 4. Honest status

- Corrected (F36 self-fire): Direction A (rate-primitive) is disfavored — it relocates the N_c$^2$ puzzle to N_c$^4$, does not resolve it.
- Favored: Direction B (two independent endpoint color sums) is the candidate; F24's "3" is a category mismatch (Sym^k degree, not color sum) and does not refute it.
- Test: are the d and s color sums independent or tied in the substrate matrix element? Independent ⍰ N_c$^2$ forced; tied ⍰ 9 is a Casey #5 composite, color reading retired. Decidable by the explicit FK color-tensored matrix element.
- Tier: F39 v0.1 diagnostic; substrate-color$^2$ source OPEN, fork sharpened to one computation; multi-week per Cal #189.

## 5. Closure

Pulling CKM Direction A showed it disfavored: making the rate the primitive relocates N_c$^2$ to N_c$^4$ rather than explaining it. Direction B (two independent color sums, one per generation-K-type endpoint) is the real candidate for $N\_c^2$ in the amplitude, and F24's earlier "3" computed symmetric-power degrees, not color sums, so it does not block B. The open question is now a single falsifiable fork — are the endpoint color sums independent or tied — decidable by the same FK color-tensored Bergman matrix element that F38 needs for $\varepsilon$. Two of my Phase 5 threads (vacuum $\varepsilon$, CKM color$^2$) now converge on one computational tool.

— Lyra, Sat 2026-06-06 12:50 EDT. F39 v0.1: CKM Direction A (rate-primitive) DISFAVORED — rate $\sin^2\theta\_C$ has $N\_c^4$, so making it primitive relocates the $N\_c^2$ puzzle (amplitude) to $N\_c^4$ (rate) rather than resolving it; corrects F36's tentative lean toward A. Direction B (two independent endpoint color sums, gen-1 d-quark color $\times$ gen-2 s-quark color, in the Bergman matrix element) is the candidate for $N\_c^2$ in the amplitude; F24's "3" computed Sym^k degree not color sum — category mismatch, does NOT refute B. Sharpened fork: are d,s color sums independent ($N\_c^2$ forced) or tied (9 = Casey #5 composite, color reading retired)? Decidable by explicit FK color-tensored matrix element — same Ch 7 tool F38 needs for $\varepsilon$. multiweek per Cal #189.